

Bob ↔ Grok Build connector

Input: Bob Functional Spec v0.1. This file is the freeze. Spec v0.1 is not the contract. Part B is the repo skeleton and test IDs. One PDF for the repo docs/ folder.

Goal is right. Architecture invents a Windows service, a private /v1 HTTP schema, a second session registry, and a leader-as-fleet-bus. Grok already ships grok -p, grok agent stdio, grok agent serve --bind 127.0.0.1, and %USERPROFILE%\groklsessions. Dual-attach to a live TUI is not a product. Freeze TUI xor Bob.

A0. Verdict

ACCEPT the problem. REJECT the custom control plane as MVP. MVP = thin adapter that shells official grok on CE-PRIORITY-DEV1, writes a disk outbox, never SendKeys. HTTP facade after one headless session survives a second prompt. MCP and legion after that.

Bob wanted	What exists	MVP freeze
Bridge service + /v1/workers	grok agent serve --bind 127.0.0.1:2419 --secret	No service until serve is insufficient
SendPrompt into live TUI	ACP session/prompt or grok -p -r SESSION	Headless/ACP only
SQLite under .groklbob-bridge	official sessions + grok sessions list	Overlay JSON for callsigns only
leader_up required	grok agent --leader (recon first)	One process per worker in MVP
Spawn N legion	N processes + N session ids	Cap 2; sticky cwd
MCP now	Bob local-exec	Phase 2

A1. Findings

S0-FAKEAPI Do not invent /v1/workers before wrapping grok

P0

Problem. Custom HTTP is a second protocol on top of ACP. You will debug the proxy instead of Form Prep.

Fix / freeze. WP1 is a PowerShell module + JSON files. Optional later: thin HTTP in front of grok agent serve.

Acceptance. Acceptance in A6 met without a Windows service.

S0-ONESHOT grok -p is not a long-lived worker

P0

Problem. Headless -p sends one prompt and exits. Polling thinking[tool_running on a dead process is fiction. Long-lived = ACP.

Fix / freeze. Two kinds: oneshot (-p json) and live (ACP). MVP Form Prep = oneshot + -r for turn 2. Live = WP3 if -r cannot resume.

Acceptance. Module documents both. Form Prep path uses oneshot.

S0-DUAL Freeze TUI xor Bob

P0

Problem. Spec already admits dual-attach may be impossible. That is the decision. Human observe = grok export or grok -r after Bob's turn ends.

Fix / freeze. Worker always --no-alt-screen. Bob never starts the pager TUI.

Acceptance. Human TUI not required. Not "both attached."

S0-YOLO --always-approve is not the Form Prep default

P0

Problem. Yolo is how UPD gets flipped and meetings get playPrompt. Permission is per profile.

Fix / freeze. formprep = --rules, no yolo. mud = yolo. teams = no AccessMedia. Never global yolo.

Acceptance. Form Prep argv has --rules and not --always-approve.

S0-MDRIVE Mapped M: + worktree Owner(-36)

P0

Problem. Already burned a day on libgit2 ownership on M:/ai*.

Fix / freeze. WP0 records safe.directory. Default no worktree. If git index dies: separate-git-dir on C:\src, files stay on M:.

Acceptance. First Form Prep worker does not pass -w.

S1-LEADER leader is optional

P1

Problem. Shared backend is not N cheap workers. Leader death kills clients.

Fix / freeze. MVP: one grok process per worker. leader_up nullable.

Acceptance. Health JSON works with leader_up=null.

S1-COST N workers = N bills and N dirty trees

P1

Problem. Two Form Prep agents on one EXECPREPLOCK races the mutex you already built.

Fix / freeze. max_workers_per_machine=2. Same cwd without force → worker_exists.

Acceptance. Second Start on busy cwd refused.

S2-PIPE Named pipe + HTTP + TCP is three transports

P2

Problem. Bob already has local-exec.

Fix / freeze. WP1–WP2: script only. No pipe listener.

Acceptance. Install creates no pipe.

A2. Frozen architecture

```
Bob --local-exec--> DEV1 powershell Invoke-BobWorker.ps1 -- official grok.exe
%USERPROFILE%\grok\sessions\          official history
%USERPROFILE%\grok\bob-bridge\
    config.json overlay.json audit.jsonl
workers\<id>\ inbox\  outbox\completion.json status.json last_result.json
```

Live ACP (WP3): one grok agent stdio (or serve --bind 127.0.0.1:2419 --secret) per live worker. Adapter holds the client. Do not also attach the TUI to that session_id.

A3. Official surfaces

Surface	Command	Use
Oneshot	grok --no-auto-update -p TEXT --cwd PATH -s ID --output-format json	MVP task
Resume	same + -r ID	Second prompt / post-reboot
List	grok sessions list (flag from WP0)	Inventory
Export	grok export ID outbox/transcript.md	Evidence
ACP stdio	grok agent stdio	WP3 live
ACP serve	grok agent serve --bind 127.0.0.1:2419 --secret	WP3.5 HTTP
Inspect	grok inspect --json --cwd PATH	Health
Forbidden	SendKeys / UIAutomation / alt-screen scrape	Hard stop

A4. Work packages

WP0 Recon — tools/Invoke-Recon.ps1 writes docs/wp0-recon.md. Do not guess flags. Freeze argv templates from the log.

WP1 Adapter — Get/Start/Send/Status/Result/Stop/Export. Oneshot spawn only.
\$env:BOB_GROK_EXE + \$env:BOB_BRIDGE_HOME.

WP2 Packets — status / completion / prompt schemas. Adapter synthesises completion if the worker does not write it. Never invent ok=true.

WP3 Live ACP — only if WP0/BT3 show -p cannot resume. Prefer official serve over a new Windows service.

WP4 Install — clone + Install-OnDev.ps1. No SCM for oneshot.

WP5 Bob skill — SKILL.md that local-execs the adapter. MCP wrappers after two Form Prep completions.

WP6 Golden path — Form Prep WhatIf → Prepare-Forms one name → Get-LastFormPrepResult → completion.json → resume WhatIf.

WP7 Phase 2 — MCP, many machines, dashboard, inbox watcher, Linux. Gate: WP6 in last-dev-run.md.

A5. Profiles

Profile	yolo	rules	max
formprep	false	DEV only. Never SQL-flip UPD. Never AllUnprepared. Never commit secrets.	1 / cwd
teams	false	No AccessMedia.All. No claiming fixture speak is audible.	1
mud	true	No spend. Stay on configured host.	2
generic	false	No production deploys. No force-push.	2

A6. Security + hard stops

- WP1 binds nothing. WP3.5 bind 127.0.0.1 + secret only.
- No XAI_API_KEY in repo, overlay, completion, audit. Reuse grok login.
- audit.jsonl: id, time, session, cwd, profile, sha256 of prompt.
- Simon-only: -Force, yolo change, cap raise, second worker same cwd.
- Hard stops: SendKeys, Service SCM before WP3 needed, second session registry, global yolo, dual-attach claim, worktree on M: without safe.directory.

Build this even if GitHub is sick

If git add dies with Invalid argument on NAS: files stay on M:, git init --separate-git-dir=C:\src\GrokBobBridge.git. Do not move the product cwd to C:.

B0. Skeleton

```
GrokBobBridge/
  README.md  LICENSE  .gitignore
  docs/      this-pdf  wp0-recon.md
  schemas/   status completion prompt-packet overlay health
  src/       BobBridge.psd1 .psm1
    Public/  Get-BobHealth Get-BobWorkers Start-BobWorker
              Send-BobPrompt Get-BobStatus Get-BobResult
              Stop-BobWorker Export-BobTranscript
    Private/ Get-BridgeRoot Invoke-Grok ConvertTo-Completion
              Test-SameCwd Write-Audit Get-Profile
  config/default.json
  tools/      Install-OnDev Invoke-Recon Test-Pack Fake-Grok
  tests/      BT0* off-DEV   BT1-BT6 DEV1   fixtures/  last-dev-run.md
```

B1. Fake-Grok

Invoke-Grok honours \$env:BOB_GROK_EXE. Test-Pack points it at tools/Fake-Grok.ps1. Never call real grok in BT0*.

```
-p PONG → JSON {sessionId, result:"PONG", stop_reason:"EndTurn"} exit 0
-p FAIL → exit 1 {stop_reason:"Error"}
-r ID -p ... → resumed=true if fake home has ID
sessions list → fake home
export ID → markdown
agent stdio → exit 2 wp3
version → fake-0.0.1
```

B2. Off-DEV tests (Test-Pack.ps1)

ID	Pass when
BT0 parse	every .ps1/.psm1/.psd1 parses
BT0b schema	five schemas JSON; required keys present
BT0c no-sendkeys	src/ has no SendKeys UIAutomation WScript.Shell
BT0d oneshot	PONG → last_result + completion ok + overlay + process gone
BT0e resume	Send-BobPrompt same id after overlay-only restart; resumed=true
BT0f cwd	second Start same cwd no -Force → worker_exists; overlay count=1
BT0g profile	WhatIfArgv has --rules, not --always-approve/--yolo
BT0h synth	FAIL → completion.failed never ok; missing result → blocked
BT0i audit	jsonl line + sha256; no XAI_API_KEY

Use temp BOB_BRIDGE_HOME. Must not touch real %USERPROFILE%\grok\bob-bridge. Must not require grok on PATH.

B3. DEV1 tests (skip off box)

ID	Pass when
BT1 health	grok_installed + logged_in or explicit not_logged_in then STOP
BT2 oneshot	PONG, session id, duration < 3 min
BT3 resume	new shell; -r in argv log; PONG2; same session id
BT4 cwd	second start refused
BT5 profile	formprep argv --rules, no yolo
BT6 golden	Form Prep completion + lastPrep evidence in last-dev-run.md; no SQL-flip; no TUI

BT3 is reboot-lite. Full reboot is optional. If real -r cannot resume an ended oneshot, WP0 records it and WP3 starts. Do not silently switch to ACP.

B4. Invoke-Grok.ps1

```
Invoke-Grok -Args @(...) → { ExitCode, Stdout, Stderr, Argv, DurationMs }
Nobody else Start-Process grok.
Timeout by profile (pong 180s, formprep 1200s). On timeout kill process tree.
WorkingDirectory = worker cwd. GROK_DISABLE_AUTOUPDATER=1. Do not set XAI_API_KEY.
```

Start-BobWorker -WhatIfArgv writes outbox/argv.txt and does not spawn (BT0g/BT5).

B5. Recon script sections

```
1 host/user/grok path/version
2 --help sessions/agent/leader
3 sessions list raw
4 inspect --json --cwd formprep (trim secrets)
5 oneshot PONG --output-format json --max-turns 1 → sessionId field names
6 -r thatId PONG2
7 dual-attach only if a TUI is already open; record error; do not start one
8 safe.directory + git status formprep
9 freeze: oneshot_argv_template, session_id_json_path, resume_flag, sessions_list_cmd, leader_required
```

B6. Failure matrix

Symptom	state	Bob
grok missing	installed=false	Stop; install
not logged in	logged_in=false	Stop; device-auth
timeout	error/failed	Kill tree
grok exit 1	done + failed	stderr 2k cap
exit 130/143	stopped	not ok
stdout not JSON	blocked	save outbox/raw.txt
Owner(-36)	failed + needs_human	do not retry -w
second cwd	worker_exists	Send-BobPrompt
OPEN formprep park	blocked	RepairOpenParks; no second worker
prompt has password=	refuse	do not write packet

B7. Schema required keys

Schema	required
health	ok, grok_installed, logged_in, grok_version, worker_count, leader_up (nullable), machine
overlay	workers[] {sessionId,title,cwd,kind,profile,createdAt}
status	sessionId, kind, state, cwd, title, updatedAt
completion	id, session, status, summary, evidence, needs_human, next_suggested
prompt-packet	id, from, to_session, goal, constraints, success, reply_channel

B8. last-dev-run.md template

```
date / computer / grok version / adapter commit
BT1 health JSON (no secrets)
BT2 sessionId duration PONG excerpt
BT3 resume argv with -r, PONG2 excerpt
BT6 formprep exit, lastPrep before→after, restoreOk, completion.status
Did not SQL-flip. Did not open TUI.
```

B9. Monday order

- Skeleton on M:\ai\GrokBobBridge (or Simon's name). separate-git-dir on C:\src if M: git dies.
- Fake-Grok + schemas + Test-Pack + BT0–BT0i green off-DEV.
- WP0 recon when grok is logged in.
- Real grok: BT1–BT3.
- BT6 Form Prep last.
- No WP3/MCP/service because git is blocked or BT3 has not failed.

B10. Disposition

Spec v0.1 is a good brief and a bad implementation brief. Wrap grok that exists. Prove Test-Pack off-DEV, then PONG/-r on DEV1, then Form Prep without paste.

Sources: x.ai/docs/build/cli/headless-scripting, cli/reference, grok-build user-guide 14–15. Bob spec v0.1. agentic_fomprep landmines same day.